

ECA0905-DIGITAL SIGNAL PROCESSING

DATE:24-07-2026

EXPERIMENT 5: Design of High Pass Filter using Windowing Techniques.

5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

PROGRAM

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---

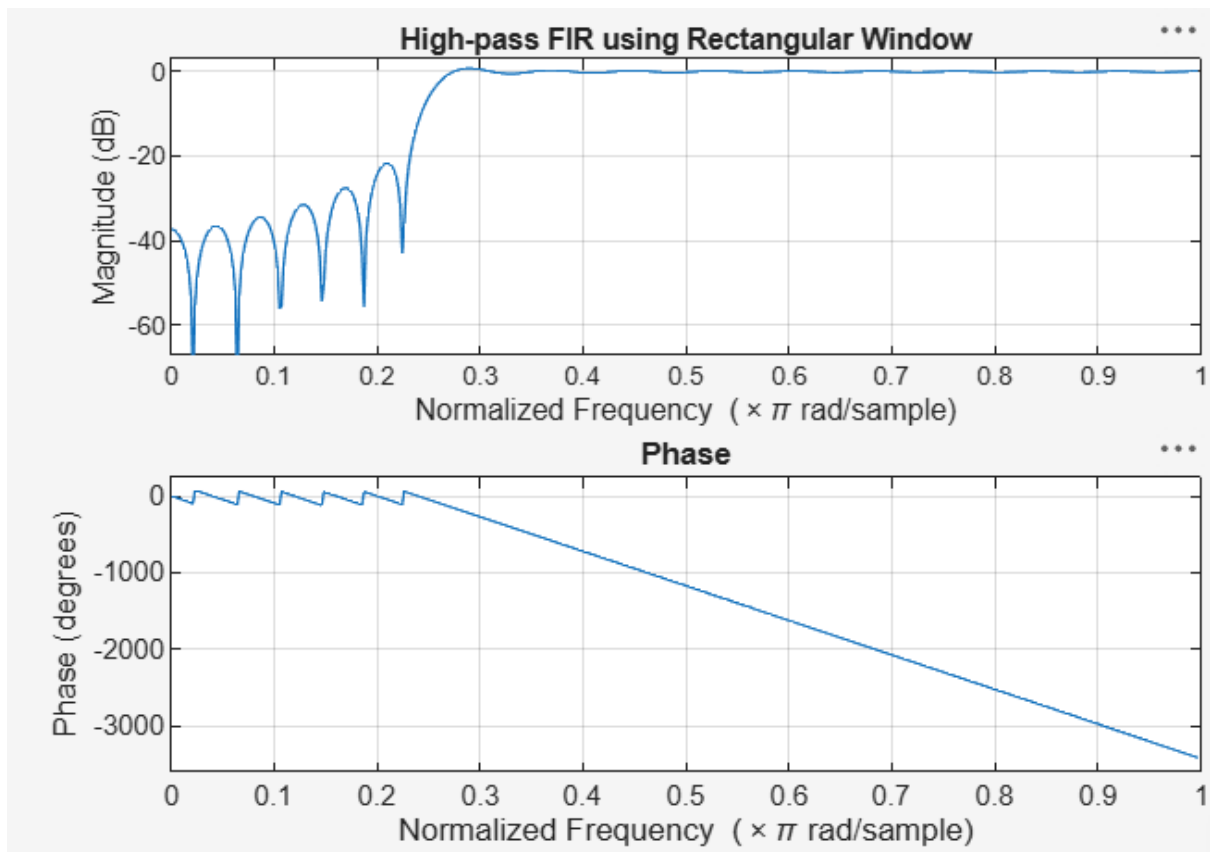
figure;
```

```

freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');

```

OUTPUT



5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

PROGRAM

```
clc;

clear;

close all;

%% Input Parameters

fp = 2000;      % High-pass cutoff frequency (Hz)
fs = 5000;      % Sampling frequency (Hz)
N = 60;         % Filter order

% Normalized cutoff frequency
wc = fp/(fs/2);

%% FIR High-Pass Filter using Hamming Window
b_hamm = fir1(N, wc, 'high', hamming(N+1));

%% FIR High-Pass Filter using Hanning (Hann) Window
b_hann = fir1(N, wc, 'high', hann(N+1));

%% Display Filter Coefficients

disp('-----');
disp('High-Pass FIR Coefficients (Hamming Window)');
disp('-----');
disp(b_hamm);
disp('-----');
disp('High-Pass FIR Coefficients (Hanning Window)');
disp('-----');
disp(b_hann);

%% Frequency Response - Hamming Window
```

```

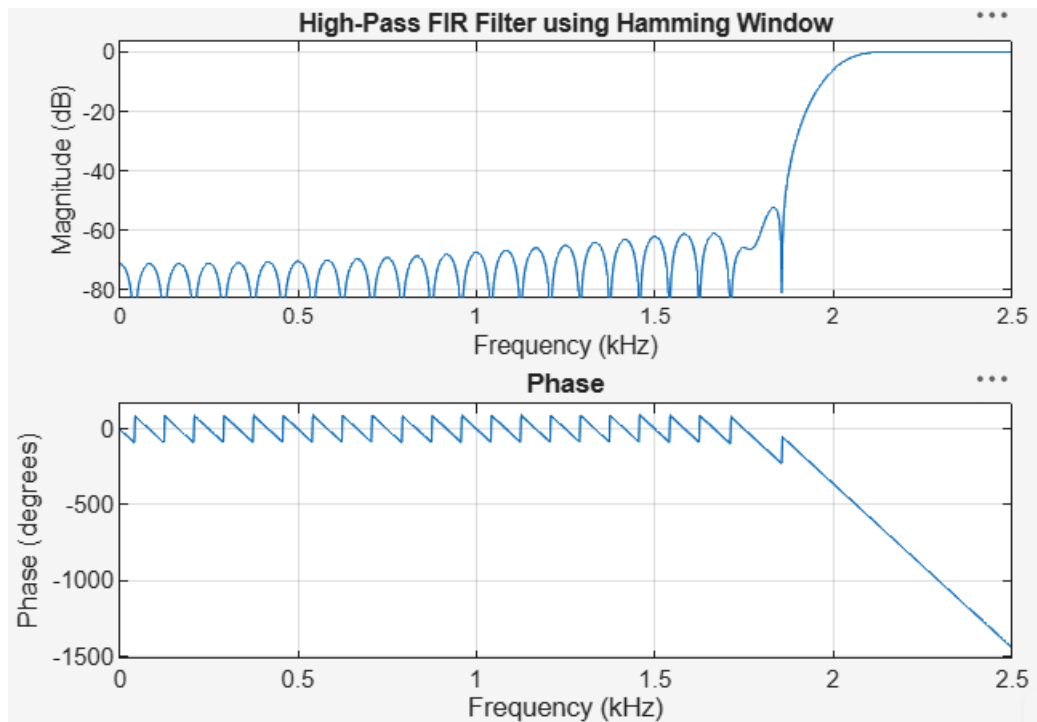
figure;
freqz(b_hamm,1,1024,fs);
title('High-Pass FIR Filter using Hamming Window');
grid on;

%% Frequency Response - Hanning Window
figure;
freqz(b_hann,1,1024,fs);
title('High-Pass FIR Filter using Hanning Window');
grid on;

%% Magnitude Response Comparison
figure;
[H1,f] = freqz(b_hamm,1,1024,fs);
[H2,~] = freqz(b_hann,1,1024,fs);
plot(f,20*log10(abs(H1)), 'b', 'LineWidth', 2);
hold on;
plot(f,20*log10(abs(H2)), 'r--', 'LineWidth', 2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Comparison of Hamming and Hanning High-Pass FIR Filters');
legend('Hamming', 'Hanning');

```

OUTPUT



5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques.

PROGRAM

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 3000; % High-pass cutoff frequency (Hz)
fs = 9000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');
```

```

disp(b_hann');
%% --- Frequency Response Plots ---
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');

```

OUTPUT

